NLP 2025
Homework 1

Multiclass Classification of Cultural Items

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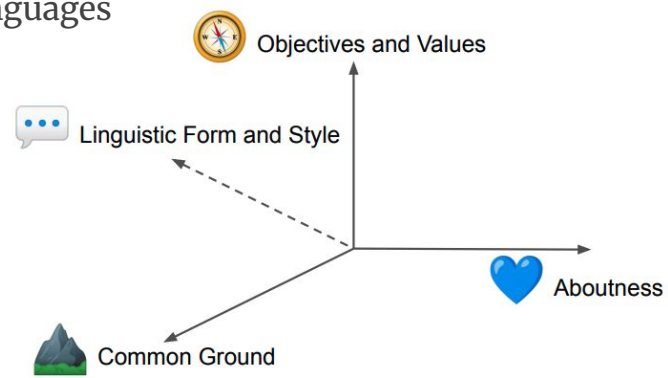
What is *culture*?

"Culture is the precipitate of cognition and communication in a human population."
- Dan Sperber

- **Linguistic form and style**
  - How things are expressed in a language
  - Different cultural groups may express things differently even within same languages
- **Common ground**
  - The shared knowledge based on how people reason and communicate
- **Aboutness**
  - What information people care to convey
  - Different cultures may choose to focus on different topics in languages
- **Objectives or values**
  - The goals a culture group strives for



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What can represent the culture?

- **Linguistic form and style**
  - Semantics similarities
- **Common ground**
  - Historical Events
  - Foods
- **Aboutness**
  - Literature works
  - Art craft
- **Objectives or values**
  - Ideologies

EVERYTHING!

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Why is culture important in NLP

Ensuring Bias Mitigation

"The bride was wearing a…"

"Long white dress and a veil"

Multicultural Generalization

Ability of a language model to accurately perform tasks across different cultural contexts.

An example: understanding proverbs

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An example of multicultural generalization

Multicultural Generalization

"Don't put all your eggs in one basket" is a common idiom that means don't put all your resources or hopes into one thing, because if it fails, you will lose everything. It is advising to diversify and spread your risks, so you have multiple options and alternatives.

Regarding your question, "The neighbor before the house," I must respectfully point out that the term "neighbor" typically refers to a person who lives near or adjacent to a specific location, such as a house or apartment. Therefore, it is not possible to have a neighbor "before" a house, as a neighbor is someone who is located in close proximity to a particular residence.

I'm glad you asked! The proverb "الجار قبل الدار" is a popular Arabic saying that translates to "The guest comes before the house." It means that hospitality and taking care of guests is of utmost importance, even if it means sacrificing one's own comfort or material possessions.

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What can represent the culture?

As you can see, culture is a complicated topic, so it would be useful if there was a system that could determine whether an item is specific to a particular culture or not!

But this is also complicated:

How would you classify pizza?

Of course it is very much tied to italian culture...

But everybody knows pizza, even outside italy... It's not something that's "exclusively" italian.

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How can we classify cultural specificity

caponata

Very cultural specific of Sicily and Italy, not widely known outside italy

Pizza

Cultural specific for Italy, but, differently from caponata, people outside italy know it! You can talk about it with an American person, and they'll understand what you're talking about

Bread

Cultural agnostic. No culture is claiming cultural ownership over "bread" and even though there are a lot of varieties of it that are culturally tied, bread itself as a concept is not.

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Our definition of Cultural Items

1. **Cultural Agnostic Item**: the item is commonly known/used worldwide and no culture claims the item
   - E.g., *bridge, potato, car, ...*
2. **Cultural Representative Item**: the item is originated in a culture and/or claimed by a culture as their own, but other cultures know/use it or have similar items
   - E.g., ***spaghetti*** *(Italian but known in Western culture),* ***hamburger*** *(originated in Germany, representative of German and American culture),* ***baozi*** *(a Chinese traditional food but also can be found world widely)*
3. **Cultural Exclusive Item**: the item is known/used only in a specific culture and it is claimed by that culture
   - E.g., ***pasta alla gricia*** *(Latium-specific pasta recipe),* ***xiaolongbao*** *(steamed bun with broth inside)*

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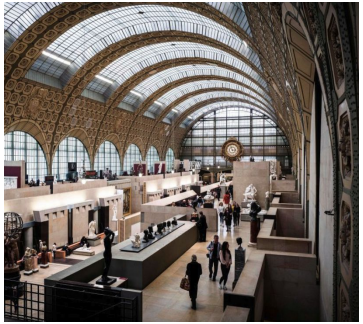
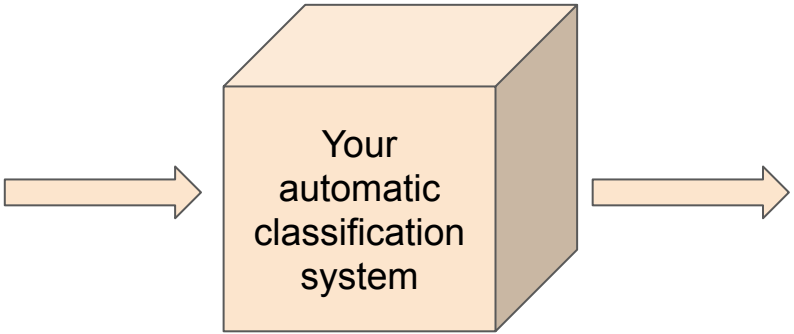
The task

Your task is to **develop an automatic classifier**, that classifies items using our taxonomy (C.A, C.R, C.E) exploring 2 approaches.

Musée d'Orsay: art museum in Paris, France





“Cultural representative”

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How will you do that?

We want you to use two different methods:

1. One LM-based method: finetune an encoder Transformer!
   a. One of BERT, RoBERTa, DeBERTa, DistilBERT, ...
2. One other method (not LM-based): you can be creative here!
   a. Word embeddings (W2V)
   b. Wikidata relationship analysis
   c. Heuristics methods / statistics on the raw Wikipedia page
   d. Use other sources of data (internet vocabularies, specific websites depending on the category, ...)
   e. ... anything else!

What data will you have?

A gold annotated dev set! Gold means manually annotated by us (you're welcome) and a **silver annotated training set**!

item	name	description	type	hypernym	classification
http://www.wikidata.org/entity/Q811389	Bauhaus Archive	museum and archive in Berlin	named entity	archive	cultural representative

As you can see this was annotated as cultural representative. We also auto generated an explanation

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What data will you have?

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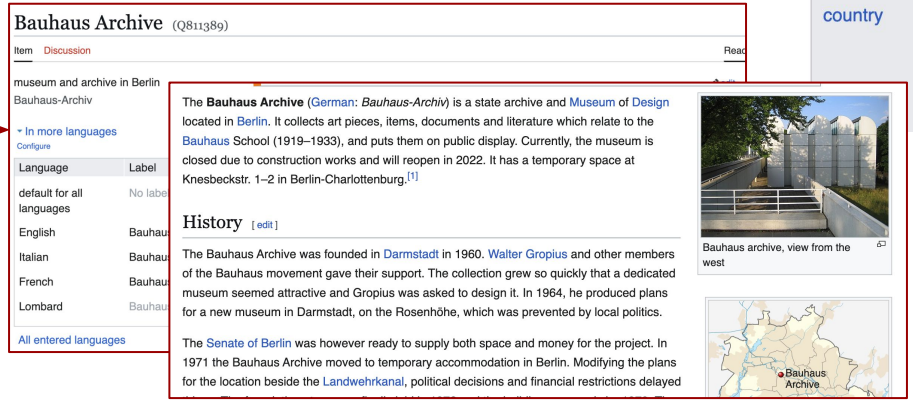

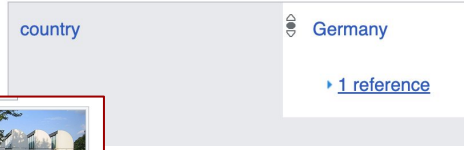

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Categories

1. Literature
2. Philosophy and Religion
3. Fashion
4. Food
5. Comics and Anime
6. Visual Arts
7. Media
8. Performing Arts
9. Biology
10. Films
11. Music
12. Sports
13. Geography
14. Architecture
15. Politics
16. History
17. Transportation
18. Gestures and Habits
19. Books

Subcategories will be released in a txt file.

Types

1. **Concept** (e.g., man, car, bridge, ...)
2. **Named Entity** (e.g., Barack Obama, the green Panda, Ponte della Musica, ...)

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Submission guidelines

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What we expect from you

This is a group assignment – you need to be in groups of 2-3 people (register here)

Mandatory - LM-based classifier

1. Train a Transformer-based model to solve a plain classification task on the provided dataset using **item_name** and **description** (you're free to add other metadata/text from Wikipedia to supplement the description)
2. Use the dev set to evaluate your model

Mandatory - non LM-based classifier

3. Devise a method to classify the culturality of entity and concepts without using LMs
4. Use the dev set to evaluate your model

Extra points for creative approaches (max +3 points)

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Where you can get the information of the Entities

- Wikipedia article texts

Spaghetti

83 languages

Article Talk Read View source View history Tools

From Wikipedia, the free encyclopedia

For other uses, see Spaghetti (disambiguation).

This article **needs additional citations for verification**. Please help improve this article by adding citations to reliable sources. Unsourced material may be challenged and removed.
Find sources: "Spaghetti" – news · newspapers · books · scholar · JSTOR *(October 2024)* *(Learn how and when to remove this message)*

Spaghetti (Italian: [spaˈgetti]) is a long, thin, solid, cylindrical pasta.[1] It is a staple food of traditional Italian cuisine.[2] Like other pasta, spaghetti is made of milled wheat, water, and sometimes enriched with vitamins and minerals. Italian spaghetti is typically made from durum-wheat semolina.[3] Usually the pasta is white because refined flour is used, but whole wheat flour may be added.[4] *Spaghettoni* is a thicker form of spaghetti, while spaghettini is a thinner form. Capellini is a very thin spaghetti, while vermicelli refers to intermediate thicknesses.

Originally, spaghetti was notably long, but shorter lengths gained in popularity during the latter half of the 20th century and now it is most commonly available in 25–30 cm (10–12 in) lengths. A variety of pasta dishes are based on it and it is frequently served with tomato sauce, meat or vegetables.

Spaghetti

Spaghetti hung to dry

Type	Pasta
Place of origin	Italy
Main ingredients	Semolina or flour, water

Cookbook: Spaghetti
Media: Spaghetti

Etymology

Spaghetti is the plural form of the Italian word *spaghetto*, which is a diminutive of *spago*, meaning 'thin string' or 'twine'.[1]

History

East Asia

See also: Chinese noodles

There are over 1,200 types of noodles commonly consumed in China today.[11] They vary widely according to the region of production, ingredients, shape or width, and manner of preparation. Due to the vast diversity of Chinese noodles, there is no single Chinese word equivalent to the Western concept of "noodles," nor is the notion of "noodles" as a unified food category recognized within Chinese cuisine.

A bowl of Bún thịt nướng

In Standard Mandarin, *miàn* (simplified Chinese: 面; traditional Chinese: 麵) means "dough" but can be used to refer to noodles made from wheat flour and grains such as millet, sorghum, and oats. While *fěn* (粉) means "powder" but can be used to refer to noodles made from other starches, particularly rice flour and mung bean starch.[12]

Wheat noodles in Japan (*udon*) were adapted from a Chinese recipe as early as the 9th century. Innovations continued, such as noodles made with buckwheat (*naengmyeon*) were developed in the Joseon Dynasty of Korea (1392–1897). Ramen noodles, based on southern Chinese noodle dishes from Guangzhou but named after the northern Chinese lamian, became common in Japan by 1900.[13][14][15][16]

Central Asia

Kesme or erişte noodles were eaten by Turkic peoples by the 13th century.

West Asia

Ash reshteh (noodles in thick soup with herbs) is one of the most popular dishes in some middle eastern countries such as Iran.

Europe

In the 1st century BCE, Horace wrote of fried sheets of dough called *lagana*.[17] However, the cooking method does not correspond to the current definition of either a fresh or dry pasta product.[18]

Italy

The first concrete information on pasta products in Italy dates back to the Etruscan civilization, the Testaroli. The first noodles will only appear much later, in the 10th or 11th centuries,[19] and there is a popular legend about Marco Polo bringing the first pasta back from China. Modern historians do not give much credibility to the story and rather believe the first noodles were imported earlier from the Arabs, in a form called *rishta*.[20] Pasta has taken on a variety of shapes, often based on regional specializations.

Jan Vermeer van Utrecht's painting of a man eating unspecified noodles (National Museum, Warsaw).

Germany

In Germany, documents dating from 1725 mention *Spätzle*. Medieval illustrations are believed to place this noodle at an even earlier date.[21]

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Where you can get the information of the Entities

- Wikipedia variants

V·T·E	Noodles	[hide]
Variants	Chinese	Biangbiang noodles · Cellophane noodles · Cumian · Jook-sing noodles · Juanfen · Knife-cut noodles · Lamian · Lai fun · Migan · Mixian · Misua · Mung bean sheets · Oil noodles · Rice noodles · Rice vermicelli · Saang mein · Shahe fen · Shrimp roe noodles · Silver needle noodles · Yi mein · Youmian
	European	Halušky · Schupfnudel · Spätzle · Pasta
	Japanese	Hiyamugi · Instant noodles · Shirataki noodles · Soba · Sōmen · Udon
	Jewish & Israeli	Egg barley · Farfel · Jewish egg noodles · Lokshen · Ptitim · Varnishkes
	Korean	*Cheonsachae* · *Dangmyeon* · *Dotori-guksu* · *Garak-guksu* · *Jjolmyeon* · *Memil-guksu* · *Somyeon* · *Sujebi*
	Thai	Khanom chin

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Where you can get the information of the Entities

- Wikidata

udon (Q471861)

type of Japanese noodle

edit

In more languages

Configure

Language	Label	Description	Also known as
default for all languages	No label defined	--	
English	udon	type of Japanese noodle	
Italian	Udon	piatto giapponese	
French	Udon	type de pâtes de la cuisine japonaise à base de farine de blé tendre	
Lombard	Udon	No description defined	

All entered languages

Statements

subclass of	Japanese noodles 0 references	edit + add reference
		+ add value

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Report Delivery

- You will write a **group report**, up to **2 pages long** excluding tables, figures and references (to be put **only at the end**).
- You should explain your **methodological choices** and **analyze** the performance of the two different approaches, both **quantitatively** (using automatic metric) and **qualitatively** (on a small random sample).
- **Delivery deadline**: *by the 4th of May, submit your homework, you must!*
- **Late submission penalty:** we will deduct 1 point for each day after the deadline, up to a maximum of -5 points.
    - **No submissions are accepted beyond 23:59 May 9th (CET)**

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Report format

- **Write 2 pages in LaTeX using the current ACL template** (available on Overleaf)
  - We will subtract points for not using this template and overflown pages
- We are interested in your reasonings and why you did X instead of Y – this must be your priority when writing the report
  - Claims need to be motivated (e.g., dataset is unbalanced → provide stats)
  - Do not explain basic concept from scratch, e.g., Transformers, BERT, LMs, etc. – we know this stuff :)
  - Keep technical details brief – your reasoning is the **priority**
  - Plots on data distribution, train metrics etc. are **more than welcome** (*at the end of the report*)
    - How to plot? Try matplotlib or seaborn
    - Don't wanna lose your head on train runs? Try wandb

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Report structure

A well-written report should ideally contain:

- **Introduction** (brief, a couple of lines)
- **Methodology** (main part of the report, explain in depth you choices)
- **Experiments** (concise)
- **Results** (reference tables and comment them, compare approaches)

  — 2 pages limit —

- **Appendix** (for Tables, Figures, Listings, Full algorithm, Bibliography)

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Code guidelines

Deliver the evaluation script as a Google Colab notebook following these guidelines:

1. **No messy, uncommented code** – it must be *readable* and *compact*

2. **File in, file out** – the eval script must accept a .csv file in input and it must output another .csv file with the predictions (*)
   - We will follow the same schema of the dev set csv.

3. **Code shall not crash** – run the whole code to be sure your code is okay

NOTE: a code failing either criteria will be **desk-rejected**.

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Code guidelines

(*) Since we are making the datasets available on the Huggingface Hub, the requirements **File in, file out** it's relaxed to accept also the test set from Hugginface (i.e., via the `datasets.load_dataset` function)

Complete usage instruction are on the Huggingface page of the dataset

Just to be clear – your evaluation script will still need to output a .csv file containing the classification labels of the test set (follow the same schema of the dataset)

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Code guidelines

IMPORTANT: Only the evaluation script should be submitted as a Google Colab notebook (i.e., the code to run your approach on the test set).

- Your other scripts (e.g., making plots, training, etc.) can be notebooks or Python scripts – you will submit them as a (public) repo / .zip file

KISS: Please keep the Google Colab as simple as possible to avoid errors and crashes

- **Ensure that we can access the Google Colab notebook** by making it shareable with anyone and assuring that you can open it from an unauthenticated/anonymous session

NOTE: a code we cannot access will be **desk-rejected.**

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Code guidelines

Remember that the code should be as most self-explanatory as possible

- Do not waste precious space on the report with the instructions to run your code!
- Instead, you can add a `README.md` containing instructions (optional)

NOTE: even if this is a group homeworks, each student is required to to be able to explain the submitted code!

- This **will be** checked at the oral exam at the end of the course.

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Evaluation

Use accuracy, precision, recall, F1-score to assess the performance of your approach.

We will set up a simple baseline (random choice) that your pipeline needs to beat.

We will also set up an online leaderboard to rank your approaches

- **Top scoring groups will get bonus points ;)**

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Evaluation

Important: we will **NOT** release the classification labels for the test set; we will run your notebook on the test set and record your scores.

- Use the dev set without labels as a reference for writing the code (it has the same structure and type of annotation of the test set)

Wikidata link	Item Name	Description	Type	Category	Classification label	Explanation
http://www.wikidata.org/entity/Q811389	Bauhaus Archive	museum and archive in Berlin	named entity	archive	-	-

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What we will reward

Creativity
This is especially important for the non LM-based classifier. **Be creative!** We would much rather see a sub-optimal creative work than a fully optimized approach with zero creativity.

Methodology
Your methodology will be thoroughly evaluated, so follow *the common machine learning best practices* (some here).

Report Clarity
The report itself is the **most important part of the homework.** You can devise the best method ever and obtain the highest scores, but if you cannot communicate well your experiments, **it will all be useless**.

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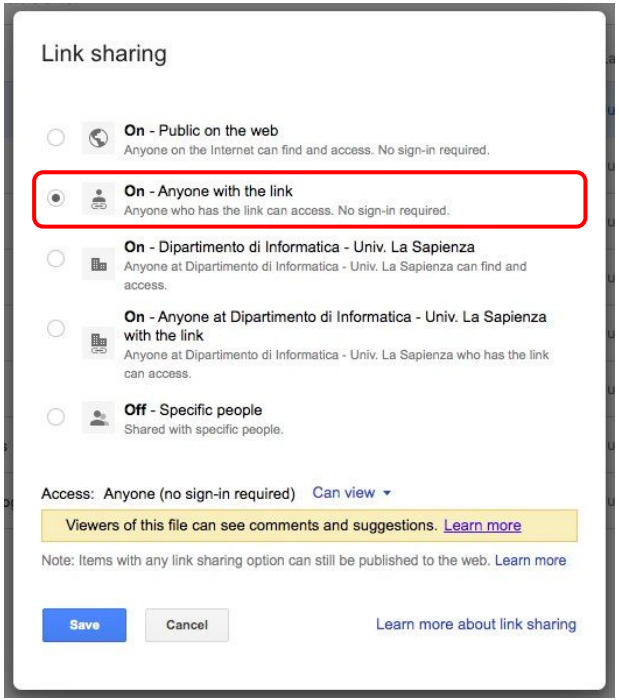
Submission Instructions



- If submitting a zip file, upload the zip on your **institutional** Drive and make it **link-shareable** and **public** to anyone.
- Make sure it is accessible via an incognito (unauthenticated) page of your browser!
- You have to submit the homework through this [submission form]() on Google Classroom. You will be asked to fill a form with the requested information and the **link** to the zip you uploaded on Drive **OR** the link to the public repo, plus a **link** to the Google Colab notebook.

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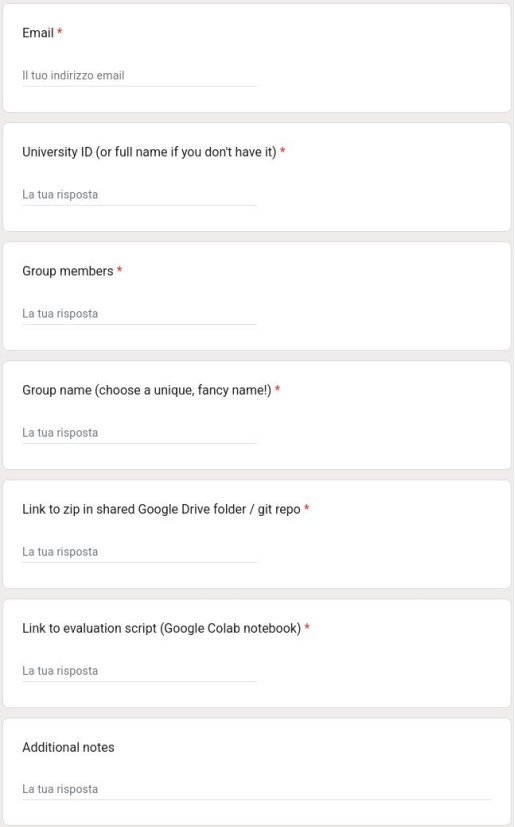
Submission instructions


Email *

Il tuo indirizzo email

University ID (or full name if you don't have it) *

La tua risposta

Group members *

La tua risposta

Group name (choose a unique, fancy name!) *

La tua risposta

Link to zip in shared Google Drive folder / git repo *

La tua risposta

Link to evaluation script (Google Colab notebook) *

La tua risposta

Additional notes

La tua risposta

When submitting using the form:

- Use your institutional email if you have one
- Please write your group members with the same spelling
- Do not abuse the additional notes field
- Please double check that the files you are sharing are accessible from anyone!

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Plagiarism

We will check for plagiarism both manually and automatically.

It is **not allowed** to:

- Copy code or reports from other students/groups;
- Share your code with other students outside your group;

Projects violating any of the above conditions will be **desk-rejected**

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Any doubts?

Use the Classroom group if you have any questions.

We **strongly** encourage you to also create a telegram/whatsapp/discord/whatever social group for easier communication between yourselves.

- Only **after you cannot solve your issues in the group**, write an email in English
- Write to **ALL the TAs** to have a higher reply rate ;)

TAs: Luca Moroni, Luca Gioffré, Lu Xu, Bruno Gatti, Alberte Fernandez Castro

Emails: moroni@diag.uniroma1.it, gioffre@diag.uniroma1.it, xu@diag.uniroma1.it, gatti@diag.uniroma1.it, castro@diag.uniroma1.it

Please start your email subjects with "**[NLP 2025 HW1]**"!

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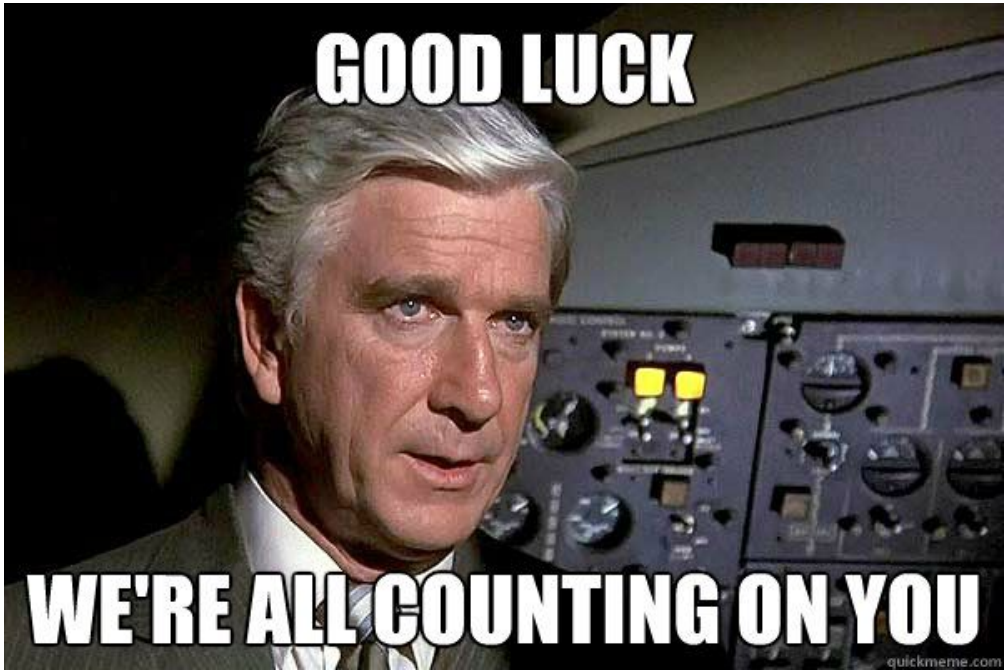

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